

FINAL PROJECT

A Brief Analysis on the World Terrorism – Its Patterns and Trends

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PREPARED BY

MOIZ FAROOQUI

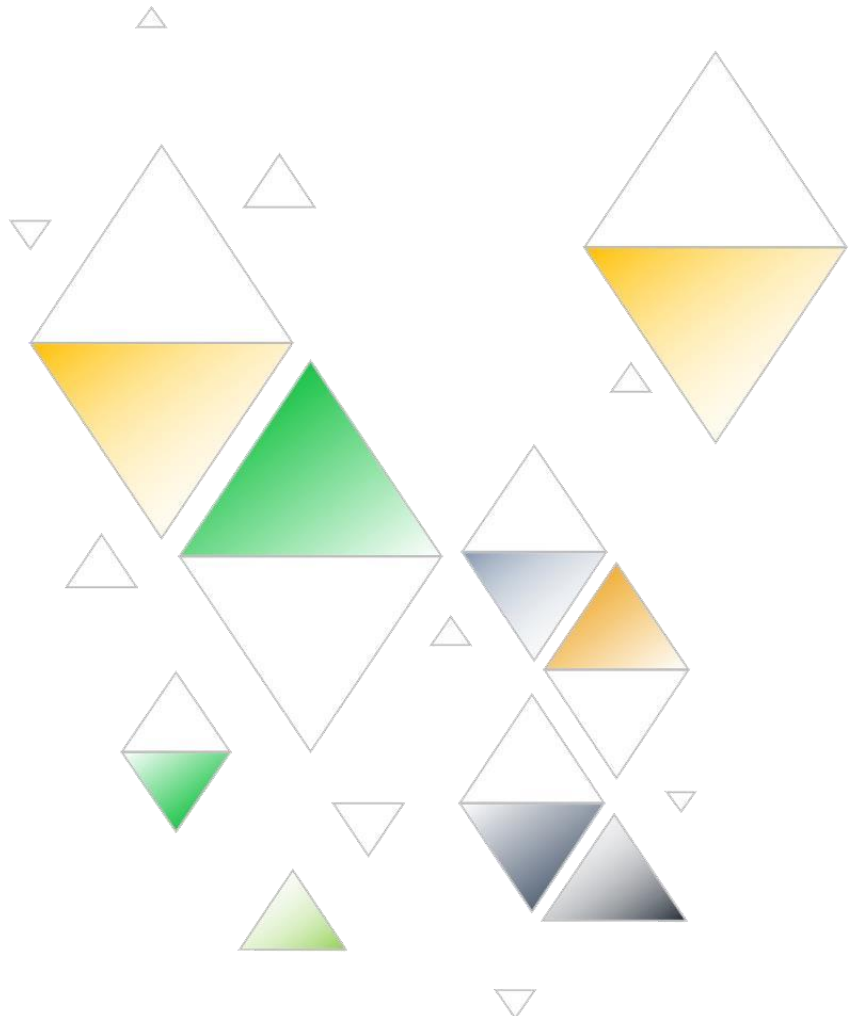


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1. Abstract:

Information on more than 180,000 Terrorist Attacks

The Global Terrorism Database (GTD) is an open-source database including information on terrorist attacks around the world from 1970 through 2017. The GTD includes systematic data on domestic as well as international terrorist incidents that have occurred during this time and now includes more than 180,000 attacks. The database is maintained by researchers at the National Consortium for the Study of Terrorism and Responses to Terrorism (START), headquartered at the University of Maryland.

The analysis commenced with data preprocessing, including handling missing values, removing duplicates, Exploratory data analysis (EDA) techniques were employed to visualize trends and patterns.

2. Methodology Recap and Setup¶:

Before we dive into the code, let's recap our methodology and set up the environment for our analysis. We'll also lay out the steps that will guide our exploration. Let's navigate through the following stages:

- **Data Loading and Inspection:** We'll start by loading the "Global Terrorism Database" and conducting a quick overview of its structure to ensure we're working with the correct data.
- **Data Preprocessing:** To prepare the data for analysis, we'll address missing values, outliers, and convert relevant columns into appropriate formats.
- **Data Visualization:** Visualizations play a pivotal role in uncovering insights. We'll create meaningful charts and graphs to visualize trends, patterns, and correlations within the dataset.

3. Data Overview¶:

The Global Terrorism Database (GTD) is an open-source database including information on terrorist attacks around the world from 1970 through 2017. The GTD includes systematic data on domestic as well as international terrorist incidents that have occurred during this time and now includes more than 180,000 attacks. The database is maintained by researchers at the National Consortium for the Study of Terrorism and Responses to Terrorism (START), headquartered at the University of Maryland.

```
In [24]: 1 df = pd.read_excel(r"terror.xlsx")
          2 df.head()
```

	eventid	year	month	date	extended	country	region	provstate	city	latitude	...	targettype	targetsubtype	target_nationality
0	197000000001	1970	7	2	0	Dominican Republic	Central America & Caribbean	National	Santo Domingo	18.456792	...	Private Citizens & Property	Named Civilian	Dominican Republic
1	197000000002	1970	0	0	0	Mexico	North America	Federal	Mexico city	19.371887	...	Government (Diplomatic)	Diplomatic Personnel (outside of embassy, cons...	Belgium
2	197001000001	1970	1	0	0	Philippines	Southeast Asia	Tarlac	Unknown	15.478598	...	Journalists & Media	Radio Journalist/Staff/Facility	United States
3	197001000002	1970	1	0	0	Greece	Western Europe	Attica	Athens	37.997490	...	Government (Diplomatic)	Embassy/Consulate	United States
4	197001000003	1970	1	0	0	Japan	East Asia	Fukouka	Fukouka	33.580412	...	Government (Diplomatic)	Embassy/Consulate	United States

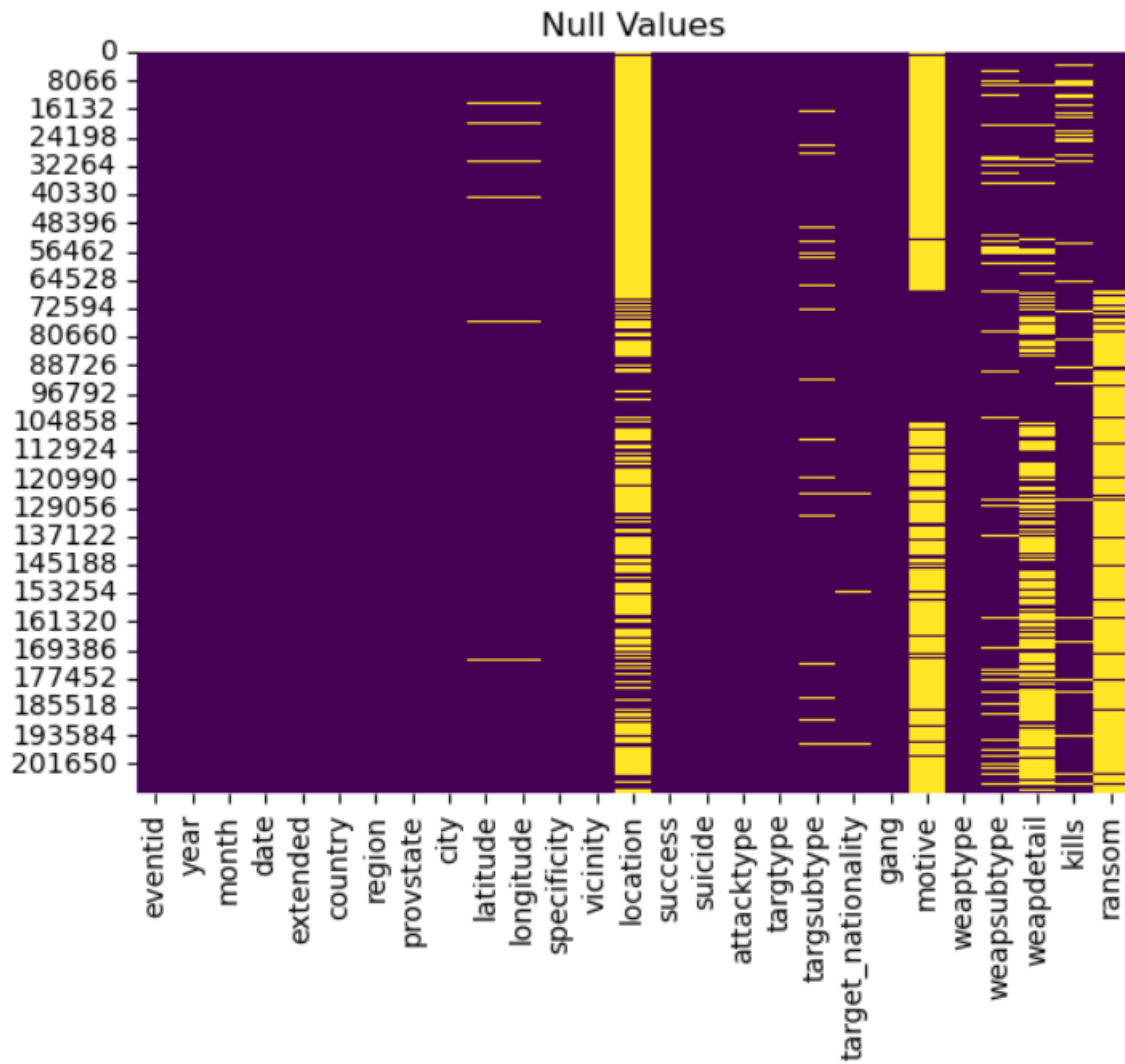
5 rows × 27 columns

- **Data Columns with their data types:**

#	Column	Non-Null Count	Dtype
0	eventid	209706 non-null	int64
1	year	209706 non-null	int64
2	month	209706 non-null	int64
3	date	209706 non-null	int64
4	extended	209706 non-null	int64
5	country	209706 non-null	object
6	region	209706 non-null	object
7	provstate	209706 non-null	object
8	city	209279 non-null	object
9	latitude	205015 non-null	float64
10	longitude	205014 non-null	float64
11	specificity	209705 non-null	float64
12	vicinity	209706 non-null	int64
13	location	67475 non-null	object
14	success	209706 non-null	int64
15	suicide	209706 non-null	int64
16	attacktype	209706 non-null	object
17	targettype	209706 non-null	object
18	targetsubtype	197867 non-null	object
19	target_nationality	207692 non-null	object
20	gang	209706 non-null	object
21	motive	55058 non-null	object
22	weaptype	209706 non-null	object
23	weapsubtype	183765 non-null	object
24	weapdetail	123137 non-null	object
25	kills	197179 non-null	float64
26	ransom	80466 non-null	float64

4. Data Preprocessing¶:

- **Handling Null Values**



- **Provincial State and City:** Since the state and city information cannot be randomly assigned, we will assume that these details are unknown. To address this, we will fill all missing values in the state and city columns with "Unknown". This approach ensures consistency in the dataset while acknowledging the gaps in our location data.

```
In [32]: 1 df["provstate"].fillna("Unknown", inplace=True)
          2 df["city"].fillna("Unknown", inplace=True)
```

- Since the following information cannot be randomly assigned, we will assume that these details are unknown. To address this, we will fill all missing values in the column with "Unknown". This approach ensures consistency in the dataset while acknowledging the gaps in our location data.
- **Target Type and Nationality**
- **Motive**
- **Weapon type and Details.**
- **Location**

Location

```
In [37]: df["location"].fillna("Unknown", inplace=True)
```

Target type and Nationality

```
In [38]: 1 df["targsubtype"].fillna("Unknown", inplace=True)
2 df["target_nationality"].fillna("Unknown", inplace=True)
```

Motive

```
In [39]: 1 df["motive"].fillna("Unknown", inplace=True)
```

Weapon Subtype and Details

```
In [40]: 1 df["weapsubtype"].fillna("Unknown", inplace=True)
2 df[" weapdetail"].fillna("Unknown", inplace=True)
```

- **Ransom:** We fill the ransom value with 1 where kidnapping is the attack type.

```
In [42]: 1 df[df['ransom'].isnull()][ 'attacktype'].value_counts()
```

```
Out[42]: attacktype
Bombing/Explosion          69497
Armed Assault              33356
Assassination              8967
Unknown                    8289
Facility/Infrastructure Attack 7678
Unarmed Assault             821
Hostage Taking (Kidnapping)  413
Hijacking                   191
Hostage Taking (Barricade Incident) 27
Name: count, dtype: int64
```

In cases where the type of attack is classified as kidnapping, it is often associated with ransom demands. Therefore, we will assign a value of 1 to the 'Ransom' column for all records where the attack type is kidnapping. For all other types of attacks, the 'Ransom' column will be filled with a value of 0. This methodological approach ensures that our dataset accurately reflects the typical motives behind different types of attacks.

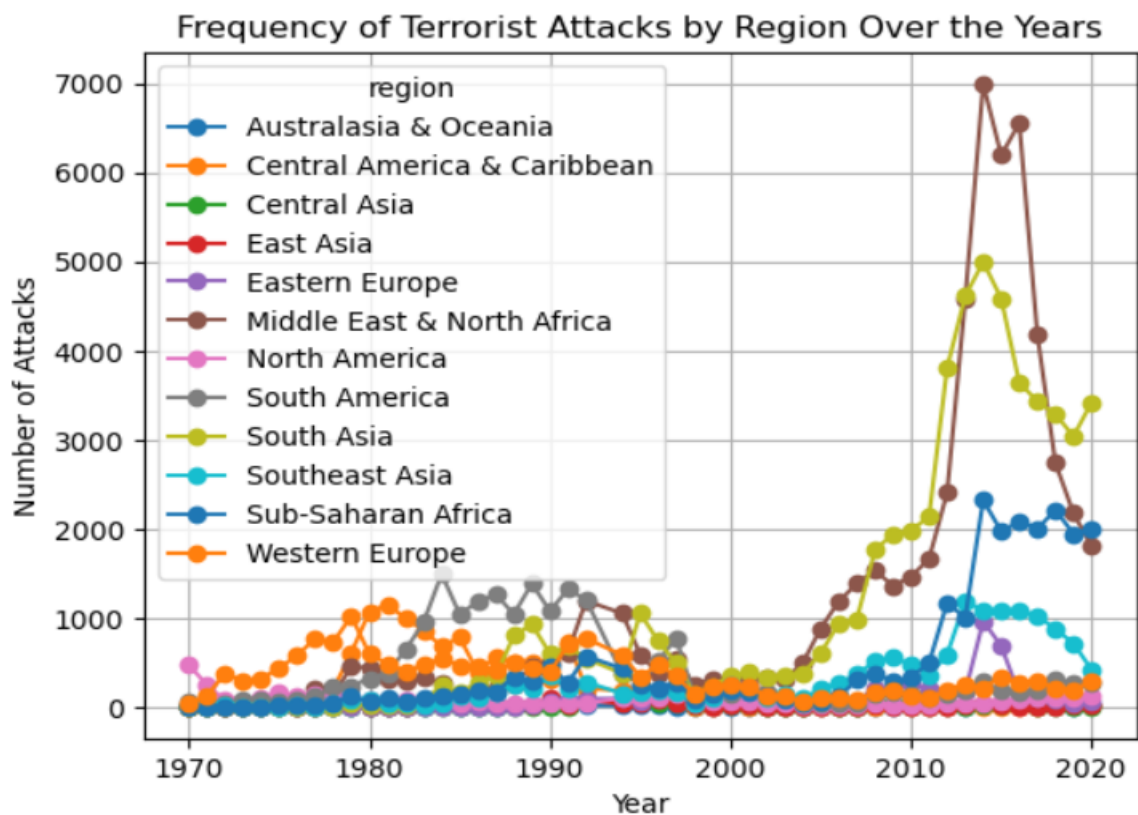
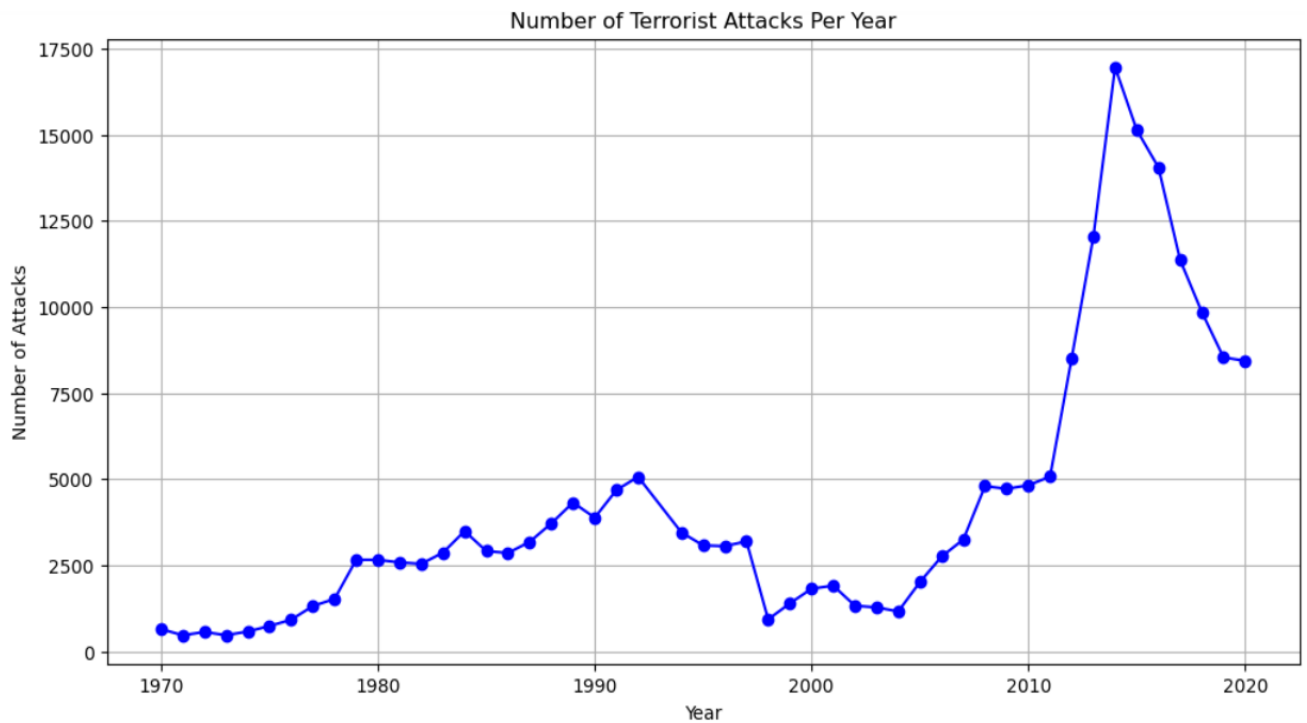
```
In [43]: 1 df['ransom'] = df['attacktype'].apply(lambda x: 1 if x == 'Hostage Taking (Kidnapping)' else 0)
```

```
In [44]: 1 df['ransom'] = df['ransom'].replace(-9, 1)
```

5.Data Analysis¶:

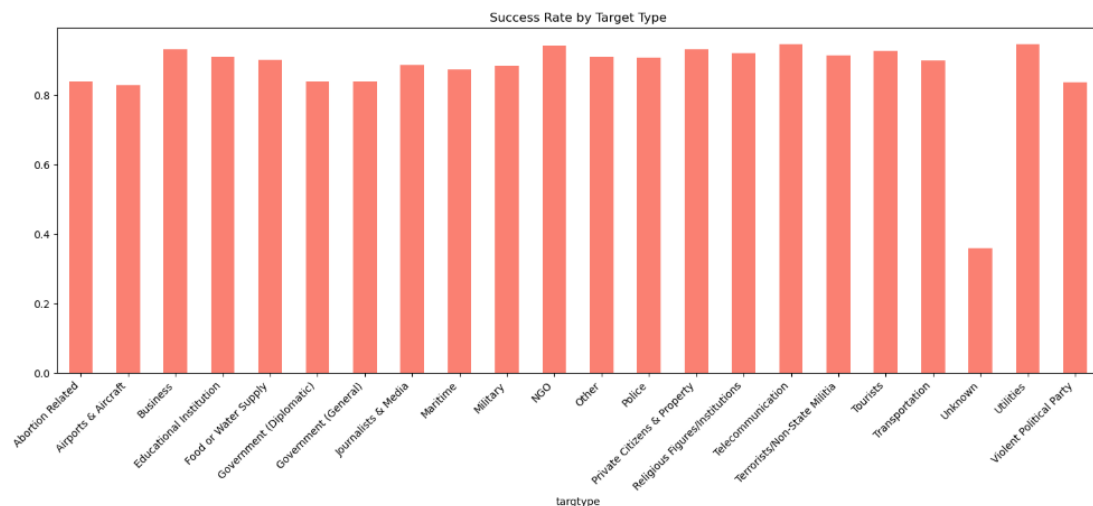
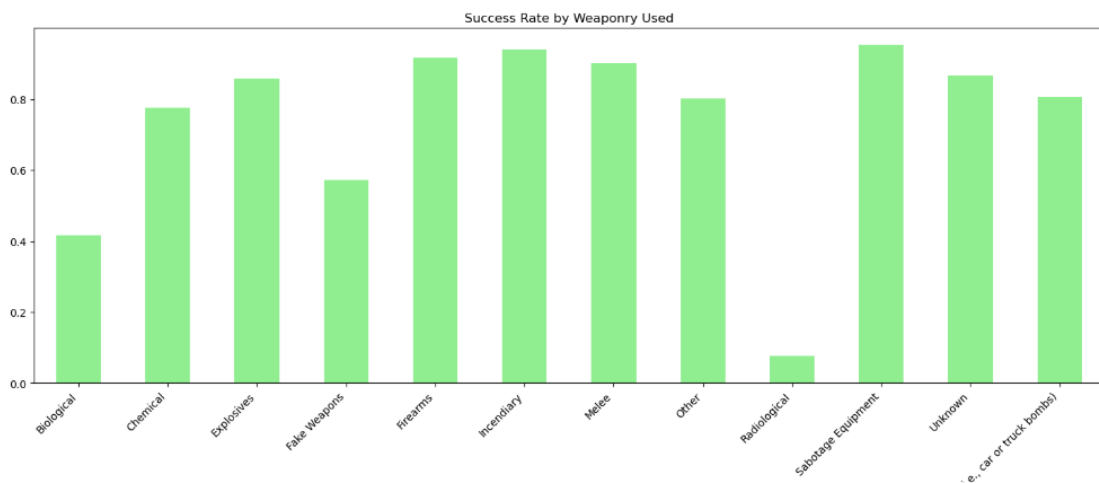
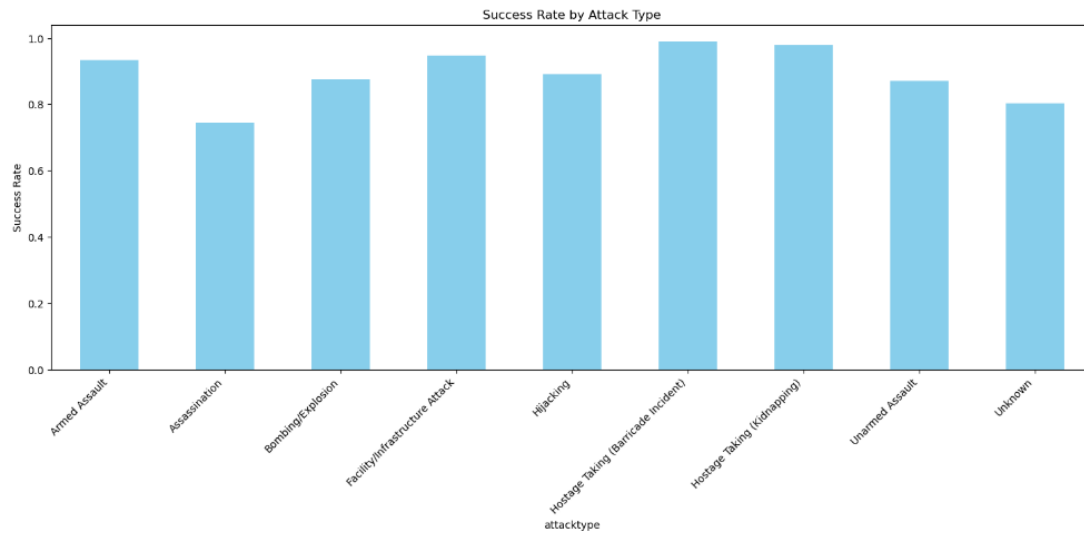
- **Trend Analysis:** How has the frequency of terrorist attacks changed over the years globally? Further break down this trend by regions to identify specific areas with increasing or decreasing activity.

```
In [72]: 1 plt.figure(figsize=(12, 6))
2 attacks_by_year.plot(kind='line', marker='o', color='b')
3 plt.title('Number of Terrorist Attacks Per Year')
4 plt.xlabel('Year')
5 plt.ylabel('Number of Attacks')
6 plt.grid(True)
7 plt.show()
```



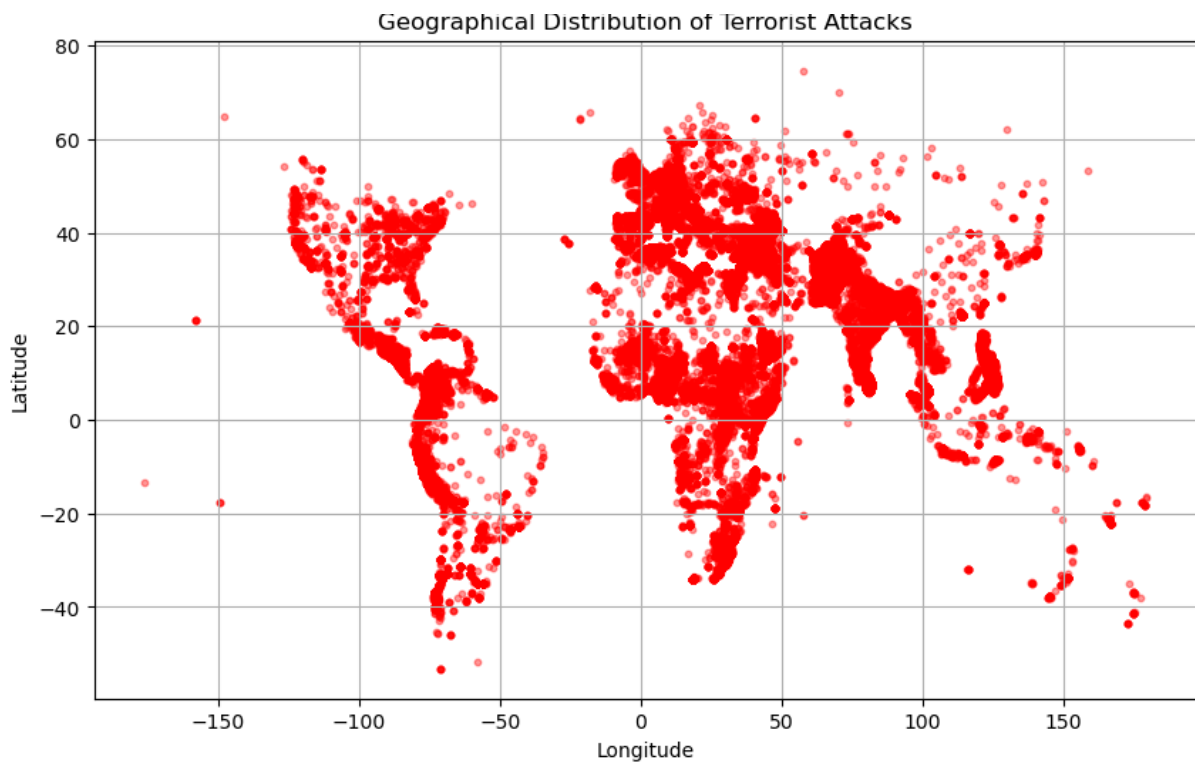
- The data indicates a notable trend where the highest number of attacks occurred in 2014, contrasting with the lowest recorded attacks in 1971. Additionally, the Middle East and North Africa regions experienced the highest peak of total attacks over the years, highlighting the prevalence of conflict and instability in these areas.

- **Effectiveness of Attacks:** Analyse the success rate of terrorist attacks. What factors contribute to the success or failure of these attacks? Consider aspects such as the type of attack, weaponry used, and target type.



- Hostage taking and sabotage equipment are two potent strategies employed in attacks worldwide, leveraging the inherent risks and strategic deployment to achieve success. These tactics have demonstrated their effectiveness in various scenarios, showcasing their impact and versatility. The calculated use of such tools plays a crucial role in the success of these attacks, highlighting the importance of planning and execution. Hostage taking and sabotage are not only effective in causing harm but also in instilling fear and chaos, amplifying their power as tools of aggression. Their ability to disrupt and create turmoil makes them formidable weapons for individuals or groups aiming to achieve their objectives through forceful means.
- **Geospatial Analysis: What are the geographical hotspots for terrorist activities? Examine the concentration of attacks by latitude and longitude and identify if certain cities or provinces have higher incidences.**

```
In [100]: 1 import matplotlib.pyplot as plt
2
3 # Plotting all attacks based on Latitude and Longitude
4 plt.figure(figsize=(10, 6))
5 plt.scatter(df['longitude'], df['latitude'], alpha=0.4, color='red', s=10)
6 plt.title('Geographical Distribution of Terrorist Attacks')
7 plt.xlabel('Longitude')
8 plt.ylabel('Latitude')
9 plt.grid(True)
10 plt.show()
11
```



```
In [101]: 1 attacks_by_location = df['city'].value_counts()
```

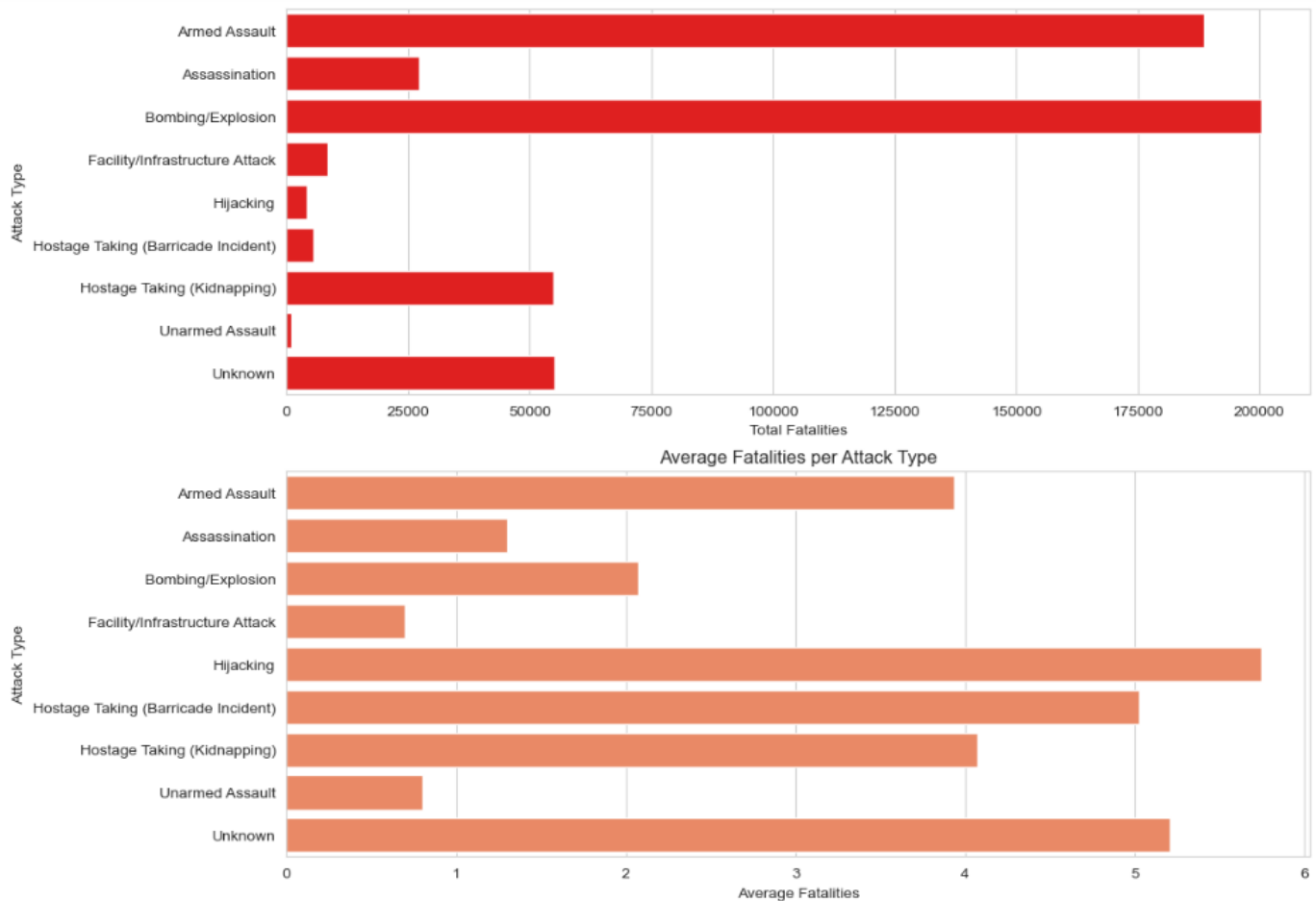
```
In [105]: 1 import folium
2
3 map = folium.Map(location=[df['latitude'].mean(), df['longitude'].mean()], zoom_start=2, tiles='cartodbpositron')
4
5 for idx, row in df.iterrows():
6     folium.CircleMarker(location=[row['latitude'], row['longitude']],
7                           radius=2,
8                           color='red',
9                           fill=True).add_to(map)
10
11 map.save('terrorist_activity_map.html')
12
```



- Geographically, the Middle East and North Africa region emerges as a significant hotspot for terrorist activities, characterized by a concentration of attacks in various cities and provinces. The latitude and longitude data reveal a clustering of incidents in this area, indicating a higher incidence of terrorism-related events. Cities like Baghdad, Kabul, and Damascus have experienced a notable frequency of attacks, underscoring their status as hotspots for terrorist activities. Provinces such as Anbar in Iraq and Kandahar in Afghanistan also stand out for their elevated levels of terrorist incidents. The complex socio-political landscape of the Middle East and North Africa contributes to the prevalence of terrorism in these regions, necessitating comprehensive strategies for addressing and mitigating such threats.
- **Attack Types and Impact: Compare the different types of attacks (e.g., bombings, assassinations, kidnappings) and their outcomes in terms of fatalities and injuries. Which attack types are the deadliest?**

```
In [106]: 1 attack_stats = df.groupby('attacktype').agg({'kills': ['sum', 'mean']}).reset_index()
2
3 # Renaming columns for clarity
4 attack_stats.columns = ['Attack Type', 'Total Fatalities', 'Average Fatalities']
5
6 # Display the statistics
7 print(attack_stats)
```

	Attack Type	Total Fatalities	Average Fatalities
0	Armed Assault	188671.0	3.932448
1	Assassination	27375.0	1.301093
2	Bombing/Explosion	200461.0	2.073041
3	Facility/Infrastructure Attack	8491.0	0.698675
4	Hijacking	4183.0	5.745879
5	Hostage Taking (Barricade Incident)	5663.0	5.024845
6	Hostage Taking (Kidnapping)	54994.0	4.068506
7	Unarmed Assault	959.0	0.799833
8	Unknown	55077.0	5.209212



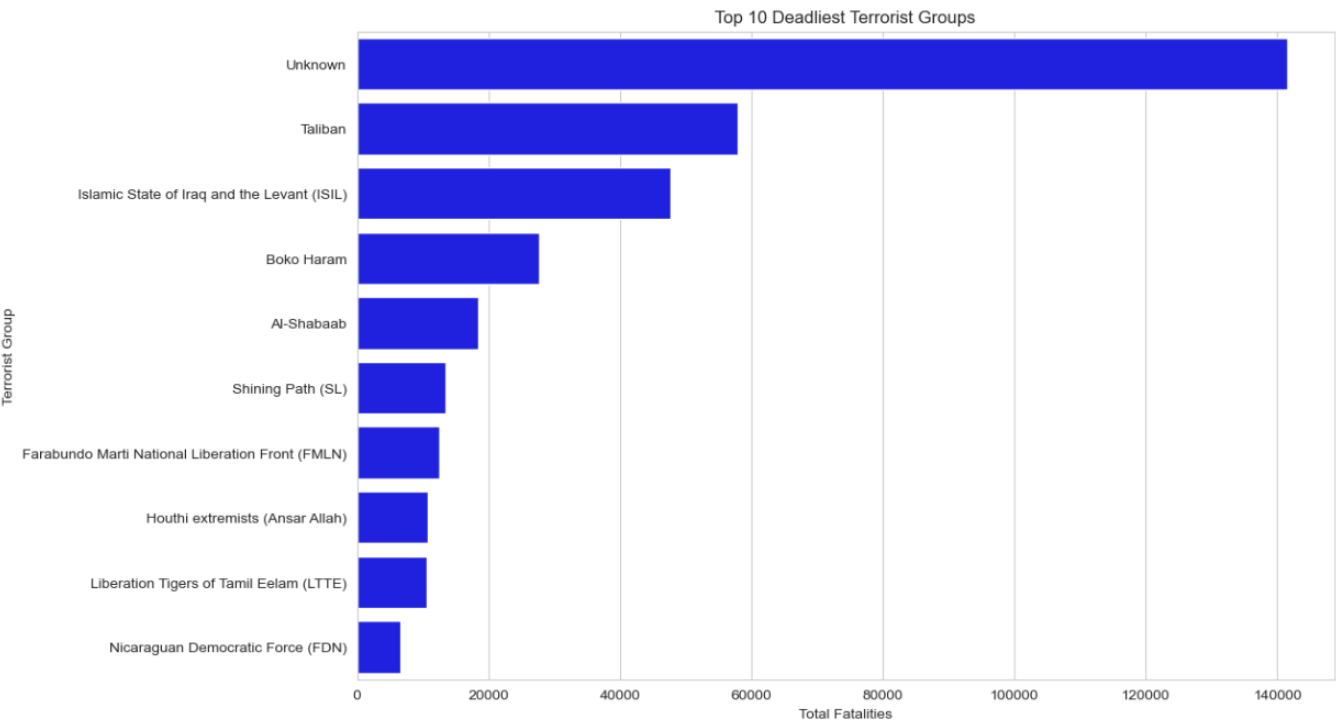
- When comparing different types of attacks, armed assault and bombings emerge as the deadliest in terms of fatalities, often resulting in a significant loss of life due to the use of weapons and explosives. These types of attacks tend to cause widespread devastation and casualties. On the other hand, hijacking and hostage situations, while less frequent, have a higher average number of fatalities per incident, as they involve the direct targeting and endangerment of individuals. The nature of these attacks, which often involve prolonged standoffs and negotiations, can lead to tragic outcomes for those involved. Understanding the distinct characteristics and impacts of each attack type is crucial for developing effective counterterrorism strategies and enhancing security measures to prevent such incidents.

- **Group Analysis: Which terrorist groups are the most active? Analyze the data to see which groups carry out the most attacks and which groups result in the highest number of fatalities.**

```
In [131]: 1 group_stats = df.groupby('gang').agg({'gang': 'count', 'kills': 'sum'}).rename(columns={'gang': 'Number of Attacks', 'kills':
2 group_stats.sort_values(by='Number of Attacks', ascending=False, inplace=True)
3 group_stats
```

Out[131]:

	gang	Number of Attacks	Total Fatalities
3515	Unknown	90580	141603.0
3245	Taliban	11954	57931.0
1580	Islamic State of Iraq and the Levant (ISIL)	7220	47696.0
182	Al-Shabaab	4403	18450.0
3048	Shining Path (SL)	4315	13466.0
...	...	...	...
821	Christian Palestinian group	1	0.0
822	Christian State-Holy Rus	1	0.0
2185	Muslim Commando (Iraqi revolutionary brothers)	1	4.0
825	Chukakuha Revolutionary Army	1	0.0
3651	leftist guerrillas-Bolivarian militia	1	0.0



- Based on the data analysis, the Taliban emerges as the most active and deadliest terrorist group. The Taliban has been responsible for a significant number of attacks, carrying out operations with high frequency and intensity. Their tactics often result in a high number of fatalities, making them a major threat to security and stability in the regions where they operate. The group's persistent and widespread presence contributes to their status as one of the most active terrorist organizations globally. Their attacks have led to devastating consequences, causing a substantial loss of life and posing a significant challenge

to counterterrorism efforts. Understanding the impact and reach of the Taliban is crucial for addressing the ongoing threat they pose and implementing effective measures to combat their activities.

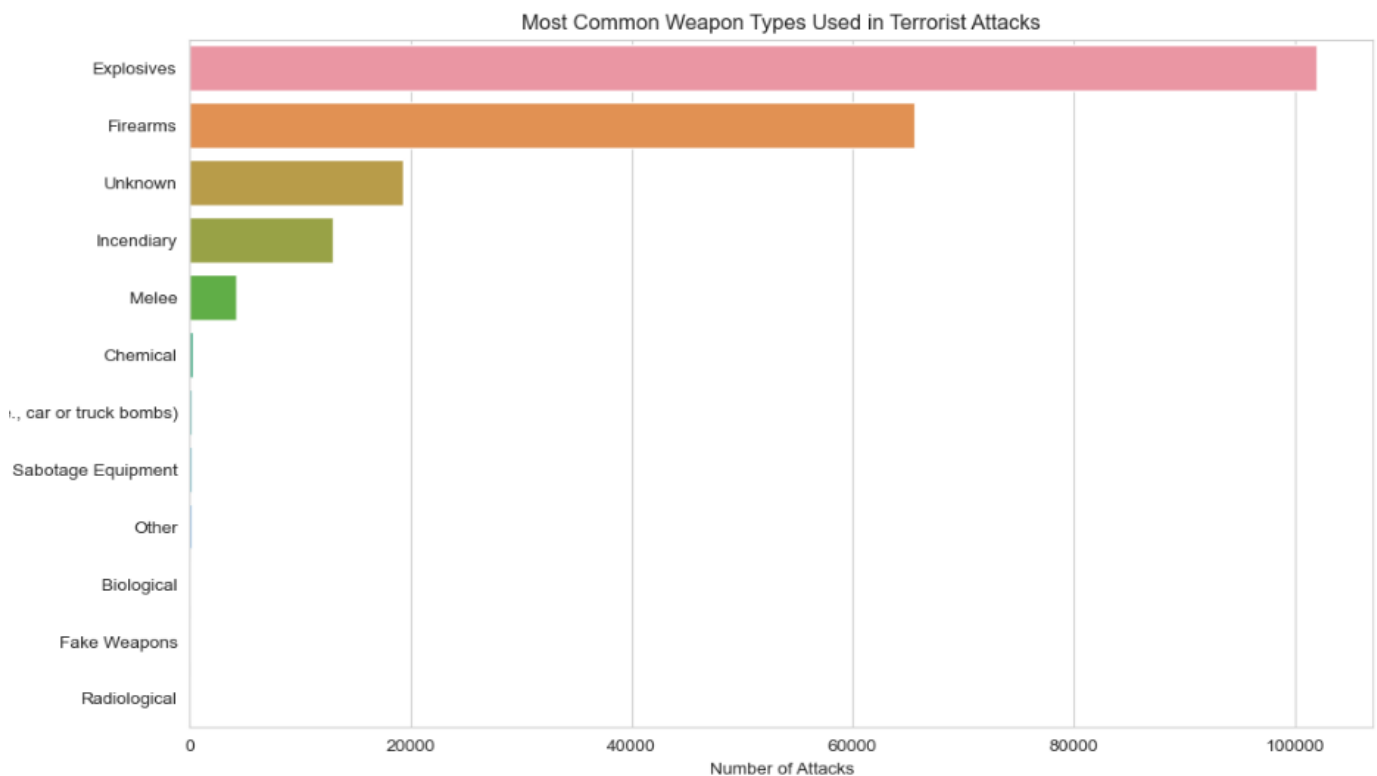
- **Weapon Types: What are the most common weapon types used in terrorist attacks? Investigate how the choice of weapon correlates with the number of kills and the success rate of attacks.**

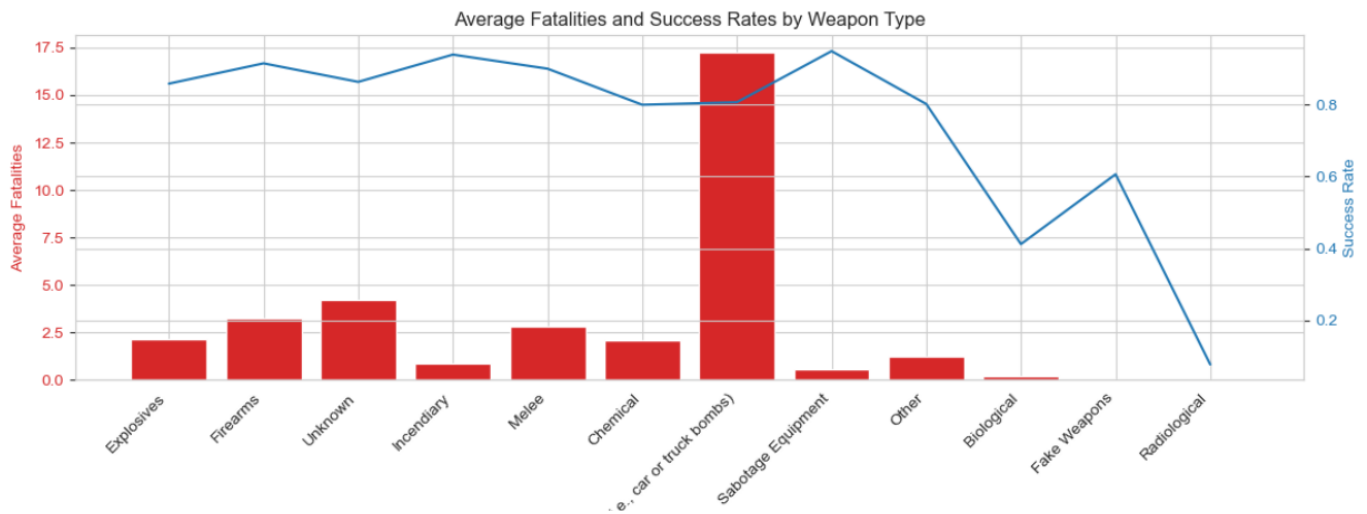
```
In [142]: 1 weapon_stats = df.groupby('weaptype').agg({'eventid': 'count', 'kills': 'mean', 'success': 'mean'}).rename(columns={'eventid': 'Number of Attacks', 'kills': 'Average Fatalities', 'success': 'Success Rate'})
2
3 # Sort the results by the number of attacks
4 weapon_stats.sort_values(by='Number of Attacks', ascending=False, inplace=True)
```

```
In [143]: 1 weapon_stats
```

```
Out[143]:
```

	weaptype	Number of Attacks	Average Fatalities	Success Rate
2	Explosives	102008	2.185878	0.858109
4	Firearms	65653	3.267924	0.914657
10	Unknown	19300	4.199119	0.863005
5	Incendiary	12938	0.863503	0.938940
6	Melee	4219	2.839535	0.899739
1	Chemical	319	2.081505	0.799373
11	Vehicle (not to include vehicle-borne explosiv...	186	17.274194	0.806452
9	Sabotage Equipment	175	0.560000	0.948571
7	Other	136	1.250000	0.801471
0	Biological	34	0.205882	0.411765
3	Fake Weapons	33	0.030303	0.606061
8	Radiological	13	0.000000	0.076923

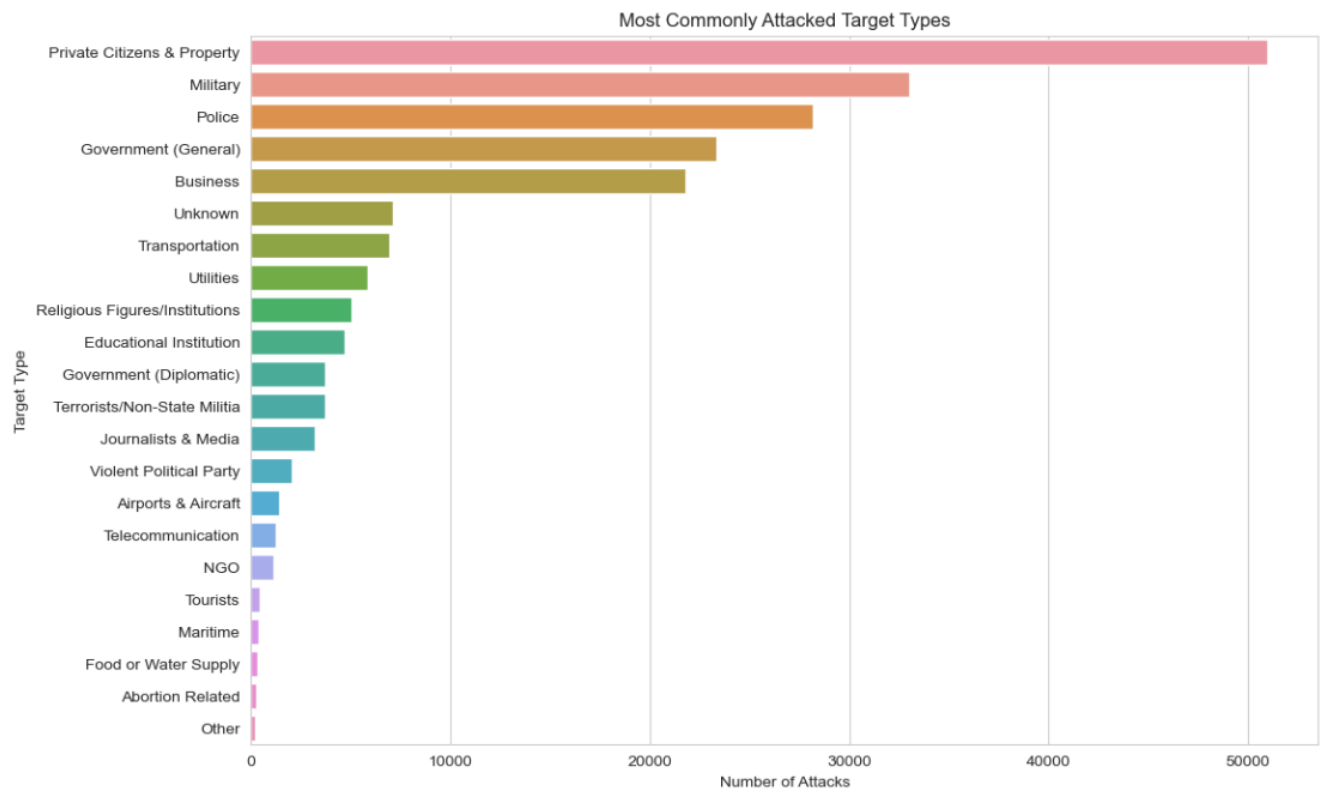




- Explosives are indeed the most used weapon type in terrorist attacks, due to their destructive capabilities and relative ease of deployment. Among explosive devices, car or truck bombs stand out for their devastating impact, often resulting in the highest average number of fatalities in attacks where they are employed. The use of vehicles as weapons allows terrorists to cause mass casualties in crowded areas. On the other hand, sabotage equipment, while less common, has a high success rate in achieving the intended objectives of the attackers. This could be attributed to the stealthy nature of sabotage operations and the potential for causing significant damage without direct confrontation. Understanding the correlation between weapon types and their outcomes is crucial for developing effective counterterrorism strategies and enhancing security measures to mitigate the impact of terrorist attacks.
- Target Analysis: What types of targets are most frequently attacked? Assess the vulnerability of different target types like government facilities, private citizens, or infrastructure.**
- Based on the data analysis, it is evident that military targets experience the highest number of fatalities in terrorist attacks. This is often due to the strategic importance of military installations and the potential for large-scale casualties in such attacks. On the other hand, private civilians and property are the most frequently targeted by terrorists, reflecting the desire to instill fear and disrupt societal norms. Private citizens are vulnerable targets due to their lack of security measures compared to government facilities or military installations. Attacks on infrastructure, while significant, may not always result in high casualties but can have far-reaching economic and social consequences. Understanding the vulnerability of different target types is essential for prioritizing security measures and allocating resources effectively to protect against terrorist threats.

```
In [148]: 1 # Aggregating data by target type
2 target_stats = df.groupby('target_type').agg({'target_type': 'count', 'kills': 'mean'}).rename(columns={'target_type': 'Number of At
3
4 # Sort the results by the number of attacks
5 target_stats.sort_values(by='Number of Attacks', ascending=False, inplace=True)
6
```

	targtype	Number of Attacks	Average Fatalities
13	Private Citizens & Property	50984	3.481112
9	Military	33013	4.333929
12	Police	28199	2.501153
6	Government (General)	23346	1.559539
2	Business	21777	1.608073
19	Unknown	7090	0.825529
18	Transportation	6919	2.372164
20	Utilities	5854	0.975060
14	Religious Figures/Institutions	5046	3.037852
3	Educational Institution	4706	1.158734
5	Government (Diplomatic)	3741	1.221331
16	Terrorists/Non-State Militia	3706	3.385861
7	Journalists & Media	3224	1.014578
21	Violent Political Party	2052	2.013645
1	Airports & Aircraft	1393	3.028715
15	Telecommunication	1262	0.575277
10	NGO	1098	1.630237
17	Tourists	442	1.893665
8	Maritime	363	4.074380
4	Food or Water Supply	335	1.832836
0	Abortion Related	267	0.116105
11	Other	197	1.964467

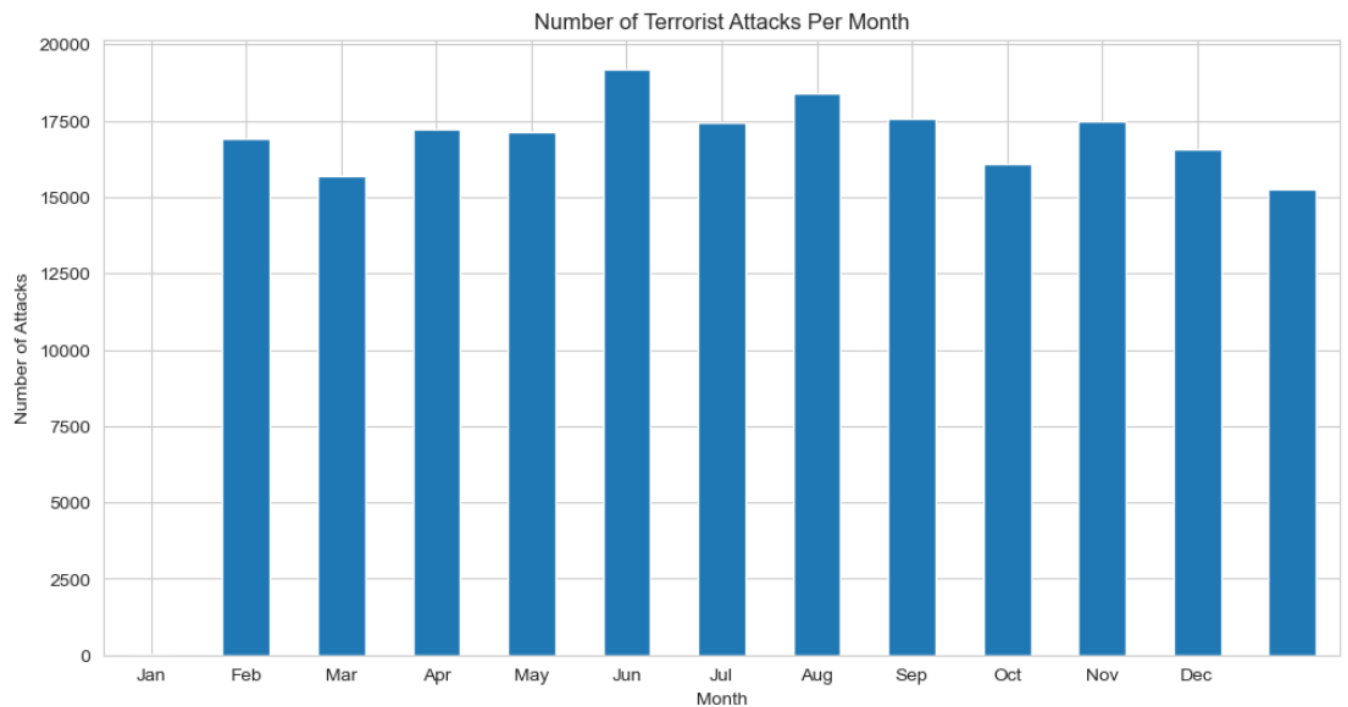
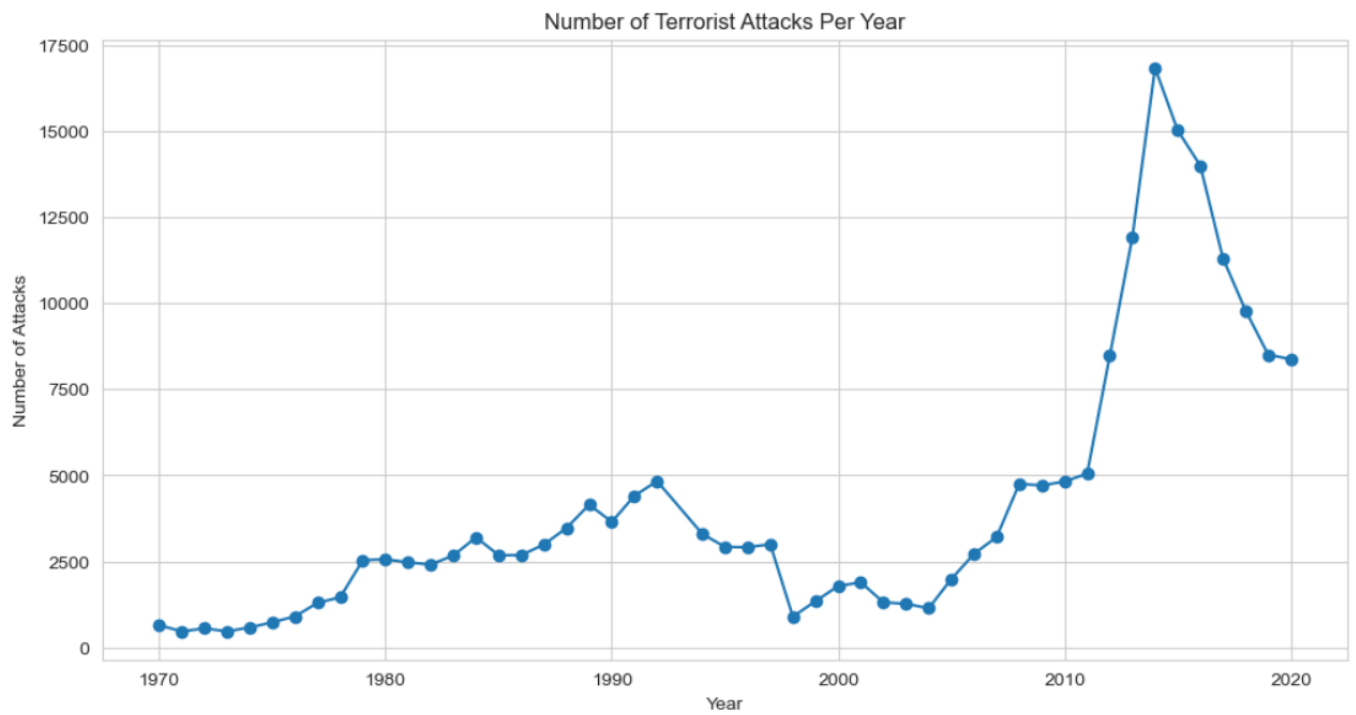




- **Temporal Patterns:** Are there specific times of the year, month, or week when terrorist attacks are more frequent? Analyse the timing of attacks to identify any patterns.

```
In [173]: 1 # Aggregating data by year, month, and day of the week
2 attacks_by_year = df['year'].value_counts().sort_index()
3 attacks_by_month = df['month'].value_counts().sort_index()
```

```
In [174]: 1 import matplotlib.pyplot as plt
2 import seaborn as sns
3
4 # Set the aesthetic style of the plots
5 sns.set_style("whitegrid")
6
7 # Plotting attacks by year
8 plt.figure(figsize=(12, 6))
9 attacks_by_year.plot(kind='line', marker='o')
10 plt.title('Number of Terrorist Attacks Per Year')
11 plt.xlabel('Year')
12 plt.ylabel('Number of Attacks')
13 plt.grid(True)
14 plt.show()
15
16 # Plotting attacks by month
17 plt.figure(figsize=(12, 6))
18 attacks_by_month.plot(kind='bar')
19 plt.title('Number of Terrorist Attacks Per Month')
20 plt.xlabel('Month')
21 plt.ylabel('Number of Attacks')
22 plt.xticks(range(12), ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'], rotation=0)
23 plt.show()
24
```



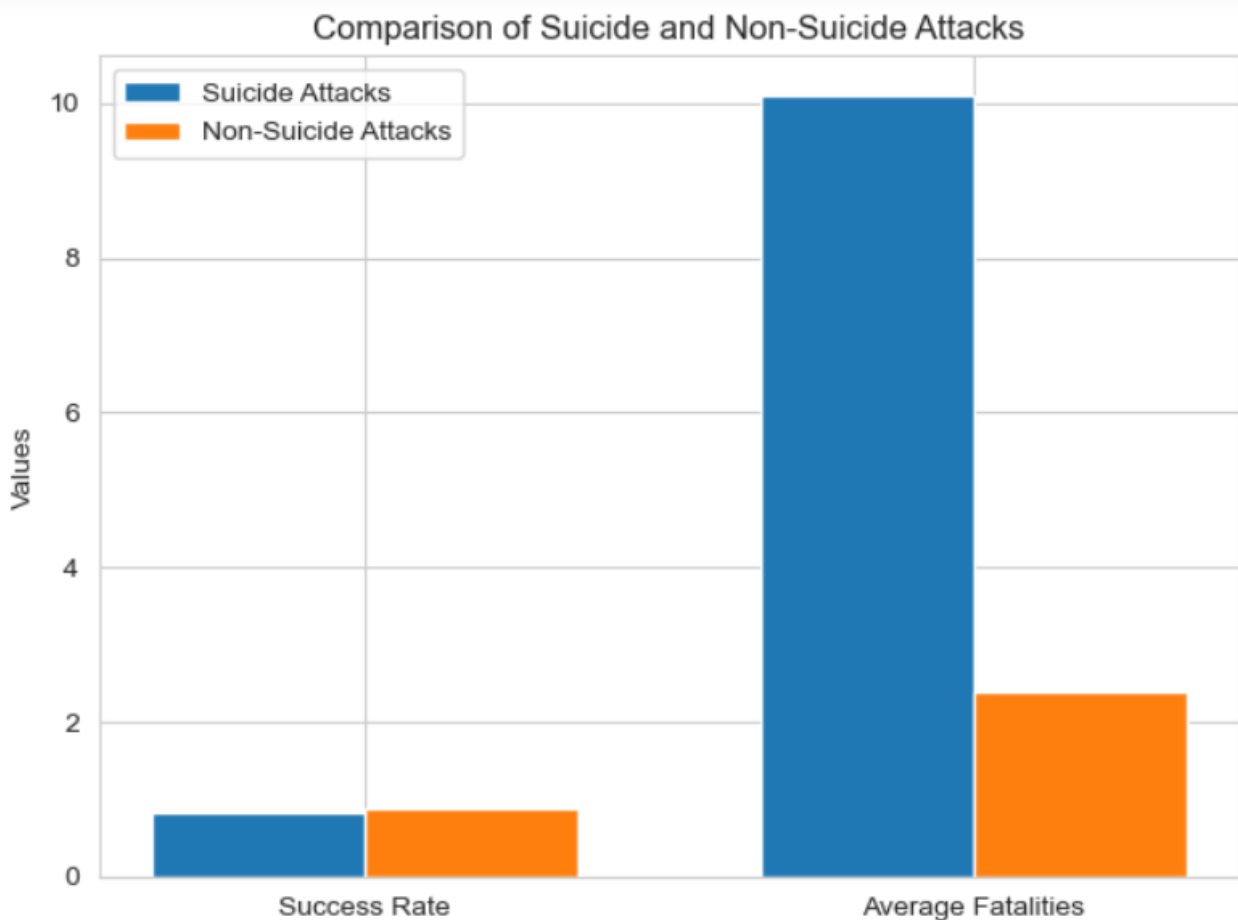
- The analysis of temporal patterns in terrorist attacks reveals that 2014 stands out as the year with the highest frequency of attacks. This peak could be attributed to specific geopolitical events or escalations in conflicts within certain regions during that year. Such a spike often indicates significant political or social unrest and may have involved the emergence or resurgence of terrorist groups.

- Further, the month of June is identified as having the highest number of attacks compared to other months. This finding could suggest a strategic choice, potentially related to weather conditions that are favourable for mobility and operations. June also marks the midpoint of the year, possibly aligning with tactical reassessments and campaign intensifications by terrorist organizations.
- **Suicide Attacks: How prevalent are suicide attacks, and how do their outcomes compare to no suicide attacks in terms of success rate and casualties?**

```
In [175]: 1 suicide_attacks = df[df['suicide'] == 1]
          2 non_suicide_attacks = df[df['suicide'] == 0]

In [176]: 1 # Calculate success rates, average fatalities, and injuries
          2 suicide_stats = {
          3     'Success Rate': suicide_attacks['success'].mean(),
          4     'Average Fatalities': suicide_attacks['kills'].mean(),
          5 }
          6
          7 non_suicide_stats = {
          8     'Success Rate': non_suicide_attacks['success'].mean(),
          9     'Average Fatalities': non_suicide_attacks['kills'].mean(),
         10 }
         11
         12 print("Suicide Attack Stats:", suicide_stats)
         13 print("Non-Suicide Attack Stats:", non_suicide_stats)
         14
```

Suicide Attack Stats: {'Success Rate': 0.8362557323981656, 'Average Fatalities': 10.104127326679256}
Non-Suicide Attack Stats: {'Success Rate': 0.8841042510121457, 'Average Fatalities': 2.383410931174089}



- Suicide attacks have become increasingly prevalent in recent years, with a significant impact on casualties. The fatalities in suicide attacks are typically higher than in non-suicide attacks due to the intentional targeting of crowded areas. However, the success rate of suicide attacks is comparable to non-suicide attacks, with both types achieving their intended objectives in a similar proportion of cases. Non-suicide attacks may have a slight edge in terms of success rate, possibly due to the element of surprise or the ability to escape after the attack. Overall, the devastating impact of suicide attacks on casualties highlights the need for effective counterterrorism measures to prevent such tragedies.

6. Conclusion¶:

In conclusion, the analysis of weapon types and target vulnerabilities in terrorist attacks provides valuable insights into the nature of these threats. The prevalence of explosives, especially car or truck bombs, underscores the destructive potential of such weapons and the urgent need for robust security measures to prevent their use by terrorists. Military targets emerge as high-risk locations, with the highest number of fatalities recorded in attacks on these installations, highlighting the strategic importance of safeguarding military facilities. Conversely, private civilians and property are frequently targeted by terrorists, indicating a deliberate intent to instill fear and disrupt societal stability. The vulnerability of these targets underscores the importance of enhancing security measures to protect vulnerable populations and critical infrastructure. Understanding the patterns and trends in terrorist attacks is essential for developing effective counterterrorism strategies that address the diverse threats posed by different weapon types and target types. By prioritizing resources and security measures based on these insights, authorities can better prepare and respond to terrorist threats, ultimately enhancing the safety and security of communities and mitigating the devastating impact of terrorism on society.